

JARDINE HILLS



Why is this project a best practice?

Challenge

The project had to provide high thermal comfort through cold winters and warm summers while sharply reducing operational energy. The main design challenge was to coordinate a high-performance envelope, efficient services, renewable energy and healthy materials without compromising the architectural concept, construction quality or everyday usability.

Solution

The response combines a 3.6 kW off-grid photovoltaic system serving the rural accommodation; solar thermal and high-efficiency heat pumps combined with an improved envelope and windows; and drought-resistant planting and drip irrigation to reduce landscape water demand. These measures were brought together through an integrated design approach and, where applicable, the Green Homes, nZEB or Passive House performance framework.

Replicability

The core package - demand reduction first, airtight construction, efficient low-temperature systems, ventilation and on-site renewables - is transferable to other Romanian homes. Replication requires climate-specific energy modelling, careful thermal-bridge and airtightness detailing, qualified installation, commissioning and clear operating guidance for residents.

Key facts and figures

Location	Mizil DJ102R, Valea Scheilor, Prahova County, Romania
Building use / function	Eco-tourism / guest cottages and existing rural house
Project type (new build, refurbishment, extension)	Extension and new-build guest accommodation
Plot area / Floor area	Approximately 270 m ² for the certified project; three new cottages plus existing buildings

Year / phase	Completed / final certification reported in 2017
Climate zone	Temperate continental, hill and vineyard influence
Thematic focus / innovation	Off-grid photovoltaic generation; solar thermal; efficient heat pumps; water-efficient landscape; rural eco-tourism
Investor / building owner	Mason Jar SRL
Architect	Not publicly disclosed in the consulted sources
Other involved stakeholders	Romania Green Building Council; renewable-energy and landscape specialists
Contact person	Mason Jar SRL / RoGBC
Further information	https://green-homes.org/property/jardine-hills/ https://www.rogbc.org/proiecte-certificate-rogbc?lang=en https://www.rogbc.org/_files/ugd/40ee3e_b5d91d7a3b71422cb98cd7a289b89c79.pdf

Jardine Hills is a extension and new-build guest accommodation project in Mizil DJ102R, Valea Scheilor, Prahova County, Romania. It functions as eco-tourism / guest cottages and existing rural house. Its main best-practice contribution is A small rural hospitality project demonstrating that off-grid renewables and water-sensitive design can be combined with local tourism. The documented strategy brings together a 3.6 kW off-grid photovoltaic system serving the rural accommodation, solar thermal and high-efficiency heat pumps combined with an improved envelope and windows, and drought-resistant planting and drip irrigation to reduce landscape water demand. Public sources provide system sizes but not a complete annual energy balance or audited carbon result. Public-source research was reviewed in July 2026; data that vary by phase are explicitly flagged in this sheet.

What was done differently?

1. Performance concept - A 3.6 kw off-grid photovoltaic system serving the rural accommodation.
2. Integrated systems and operation - Solar thermal and high-efficiency heat pumps combined with an improved envelope and windows.
3. Wider value - Drought-resistant planting and drip irrigation to reduce landscape water demand.

What was achieved?

Impact

Environmental impact: The project reduces energy and associated emissions primarily by lowering demand and then using efficient or renewable systems. Its material, landscape, waste or mobility measures add benefits beyond operational energy where documented. A verified whole-life carbon assessment was not publicly available, so the impact is described qualitatively unless a source provides a specific figure.

Economic impact: Lower energy demand can reduce utility exposure and improve long-term operating-cost predictability. Standardised high-performance specifications and third-party certification can also support quality assurance, market value and access to green-finance products. Public sources do not provide a complete investment, payback or life-cycle-cost analysis for this project.

Social impact: The main social benefits are stable indoor temperatures, better air quality, daylight and acoustic comfort, together with safer and more attractive shared or outdoor spaces where these form part of the project. The project also raises market awareness by making high-performance housing visible to residents, designers, contractors and investors.

Key Learnings

Success factors: Key success factors are an integrated design process, a clear performance framework, early coordination of envelope and building services, selection of competent suppliers, quality control during construction and commissioning. Transparent documentation and communication with occupants are essential to preserve the intended performance in use.

Lessons for replication: Replication should start with an energy and comfort target, not a catalogue of technologies. Design teams should verify thermal bridges, airtightness, ventilation, controls and renewable sizing together; record assumptions by project phase; commission systems; and monitor early operation. Where public figures differ by phase, the final as-built dataset must become the single source of truth.